

SQL Control Architecture (DDL, DML & TCL)

Complete Reference Guide & Difference Table

1. Data Definition Language (DDL)

Defines and alters the **structure/schema** of database objects. Auto-committed (cannot be rolled back).

Command	Description	Syntax Example
CREATE	Creates a new database object.	CREATE TABLE Student (...);
ALTER	Modifies existing table structure.	ALTER TABLE Student ADD Age INT;
TRUNCATE	Removes all records; keeps table structure.	TRUNCATE TABLE Student;
DROP	Permanently deletes table and structure.	DROP TABLE Student;

– DDL Example:

```
CREATE TABLE Student (ID INT PRIMARY KEY, Name VARCHAR(50));
ALTER TABLE Student ADD Email VARCHAR(100);
```

2. Data Manipulation Language (DML)

Manages and modifies **data rows/records** inside existing tables. Not auto-committed.

Command	Description	Syntax Example
INSERT	Adds new data rows into a table.	INSERT INTO Student VALUES (...);
UPDATE	Modifies existing record values.	UPDATE Student SET ... WHERE ...;
DELETE	Removes specific data rows.	DELETE FROM Student WHERE ...;

– DML Example:

```
INSERT INTO Student VALUES (101, 'Rahul');
UPDATE Student SET Name = 'Alex' WHERE ID = 101;
```

3. Transaction Control Language (TCL)

Manages transactions executed by DML statements to ensure data safety and integrity.

Command	Description	Syntax Example
COMMIT	Saves changes permanently.	COMMIT;
ROLLBACK	Undoes changes back to last commit/savepoint.	ROLLBACK;
SAVEPOINT	Sets a checkpoint within a transaction.	SAVEPOINT Check1;

– TCL Example:

```
UPDATE Accounts SET Balance = Balance - 1000 WHERE AccNo = 101;
SAVEPOINT Step1;
COMMIT;
```

4. Difference Table: DDL vs DML vs TCL

Feature	DDL (Structure)	DML (Data Rows)	TCL (Transaction)
Primary Purpose	Defines database schema/objects	Manages & modifies data records	Manages transaction stability
Target Scope	Tables, Views, Indexes	Data rows inside tables	Pending DML operations
Commands	CREATE, ALTER, DROP, TRUNCATE	INSERT, UPDATE, DELETE	COMMIT, ROLLBACK, SAVEPOINT
Auto-Commit	Yes (Saved automatically)	No (Requires explicit commit)	N/A (Executes control)
Rollback	Cannot be rolled back	Can be rolled back	Controls rollback operation
WHERE Clause	Not Supported	Supported	Not Applicable